🚀 RESEARCH PRESENTATION • ADVANCED AI AGENT SYSTEMS

MCP Orchestrator: Dynamic Tool Retrieval & Context-Aware Ranking Gateway

Resolving Context Window Saturation, Tool Confusion, and Token Inflation in Autonomous LLM Coding Agents via Just-In-Time Meta-Tool Discovery.

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PROJECT GUIDE

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CONTEXT & PROBLEM STATEMENT

Slide 2 — Introduction: The Context Bloat Crisis

PROBLEM ANALYSIS

⚠️ The "Load-All" Anti-Pattern in MCP

The **Model Context Protocol (MCP)** standardizes how AI coding agents connect to external tools. However, current client implementations rely on an unscalable naive approach:

- **Upfront Static Preloading:** Every connected server's complete JSON Schema is dumped directly into the system prompt upon initialization.
- **Explosive Token Overhead:** Connecting 10–20 downstream servers injects **21,800+ tokens** before the user types a single prompt.
- **Stateless Re-transmission:** This 21.8k schema payload is re-transmitted on *every single turn and sub-step*.

45.0k Context - 23.2k Base = 21.8k Wasted Schemas

Empirically verified: MCP tools account for nearly 50% of the entire prompt window!

⚡ Triple Failure Modes & Gateway Solution

1. Tool Confusion & "Lost in the Middle":
When faced with 60+ tools, LLMs pick wrong tools and hallucinate arguments.

2. Rapid Context Exhaustion:
Forces early conversation compaction, wiping out vital debugging logs and diffs.

3. The Solution: Dynamic Meta-Tool Gateway:
JIT retrieval exposing only `discover_tools` and `invoke_tool`.

65,400
TOKENS WASTED (3 STEPS)

3,751
WITH MCP ORCHESTRATOR

RESEARCH GOALS

Slide 3 — Research & Engineering Objectives

CORE GOALS

1. 85%–90%+ Context Reduction

Eliminate upfront tool schema bloat by replacing static registration with dynamic Top-K slicing (K=5).

Target: <750 Tokens
Down from 7,450 – 21,800

2. Context-Aware 3-Term Ranking

Design a mathematically normalized scoring function combining Query Intent, Workspace Manifests, and Past Success.

Formula: $w_1 \cdot Q + w_2 \cdot P + w_3 \cdot S$
Bayesian Empirical Reliability

3. Universal Compatibility

Non-invasive FastMCP proxy gateway with zero code modifications needed in Claude Code, OpenCode, or Cursor.

2 Meta-Tools Exposed
discover_tools & invoke_tool

4. Low Latency & Failover

Maintain end-to-end algorithmic retrieval latency under 70ms on CPU with automatic SQLite & Lexical failovers.

Latency: <67.16 ms
Background model warmup

BOUNDARIES & CONSTRAINTS

Slide 4 — Scope of the Study

SYSTEM BOUNDARIES

In-Scope System Boundaries

- **Protocol Conformance:** Complete support for Anthropic Model Context Protocol specification across stdio and SSE transports.
- **Multi-Server Ecosystems:** Seamlessly manages 8 to 20 concurrent downstream tool servers (GitHub, Filesystem, Playwright, Memory, SQLite).
- **Tool Schema Layer:** Explicit mathematical isolation and optimization of MCP Tool Schema overhead across single-turn and multi-turn workflows.
- **Context Awareness:** Workspace manifest parsing (extracting dependencies from package.json, pyproject.toml, requirements.txt).
- **Transparent Hot-Reloading:** Automated file modification timestamp (mtime) polling for downstream configuration changes.

Assumptions, Limits & Baselines

- **Fixed Baseline Separation:** Isolates MCP tool schemas from client-native overhead (e.g. OpenCode's 148 skills catalog).
- **Quantized CPU Inference:** Lightweight dense embeddings (all-MiniLM-L6-v2, 384 dim) eliminating GPU infrastructure requirements.
- **Deterministic State Machine:** LangGraph pipeline prevents infinite recursion or nondeterministic cycle loops.

Key Scope Takeaway:

MCP Orchestrator does not rewrite the agent's reasoning engine; it acts as an intelligent middleware filter that shields the LLM from schema bloat.

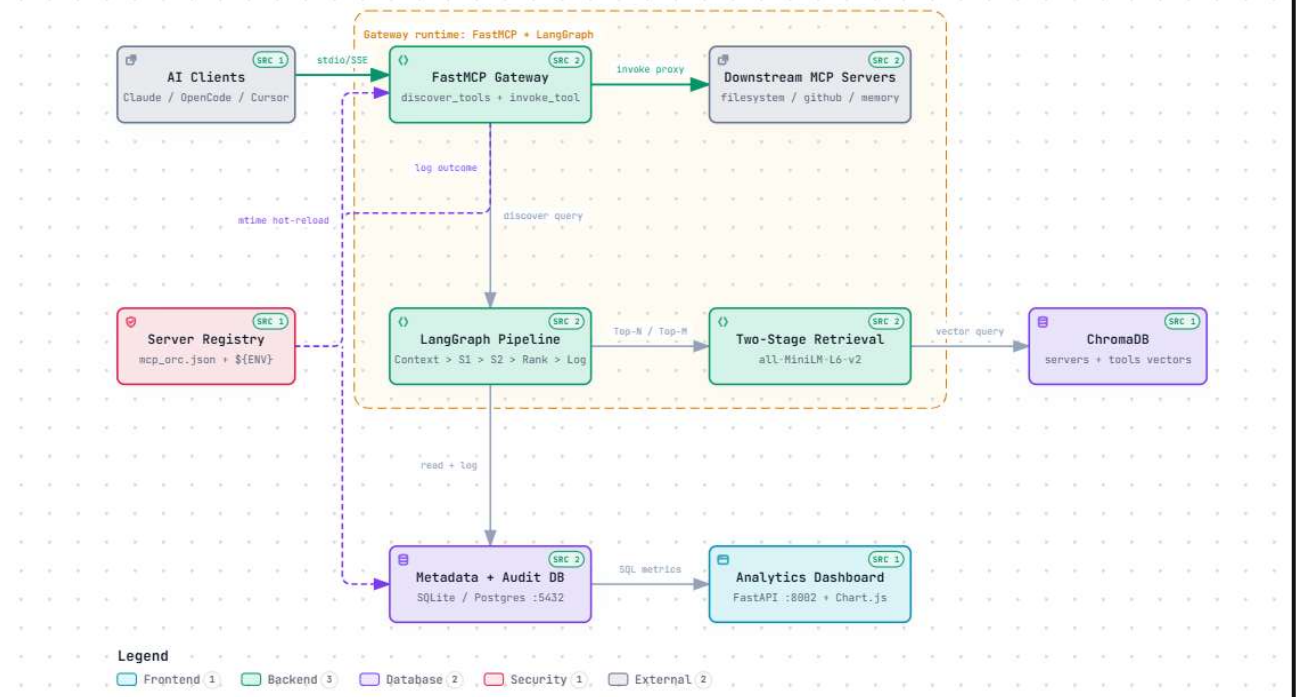
ARCHITECTURE & PIPELINE

Slide 5 — System Architecture & Methodology

SYSTEM ARCHITECTURE

End-to-End Orchestrator Architecture

FastMCP + LangGraph



AI Clients ➔ Gateway ➔ Downstream Servers

ChromaDB Vector + PostgreSQL Audit

Hierarchical Two-Stage Retrieval

Stage 1 (Server Pruning): ChromaDB `servers` collection extracts Top-N (N=3) candidate servers via query vector.
Stage 2 (Metadata Constrained): Evaluates `tools` collection filtering: `WHERE server_id IN [candidates]`.
Fallback: Automatically falls back to lexical TF-IDF if Chroma is unavailable.

3-Term Context-Aware Ranking

$$Score(t) = w_1 \cdot Q + w_2 \cdot P + w_3 \cdot S$$

w_1=0.25 (Query), w_2=0.70 (Project), w_3=0.05 (Success)

- **QuerySim:** Cosine similarity between query and tool vector.
- **ProjMatch:** Semantic match with project dependencies.
- **PastSuccess:** Bayesian reliability (Success + 1)/(Total + 2).

```
Threshold: If max(QuerySim) < 0.15, return [] to save tokens.
```

MCP Orchestrator Presentation | Architecture Diagram | ChromaDB | LangGraph

Slide 05 / 08

EMPIRICAL RESULTS

Slide 6 — Experimental Findings & Benchmarks

EVALUATION & METRICS

90.3%
CONTEXT REDUCTION
7,450.0 → 725.1 tokens (K=5)

85.0%
SUCCESS PRESERVED
Zero capability degradation

67.16 ms
RANKING LATENCY
100% CPU execution

94.3%
MULTI-STEP SAVINGS
65.4k → 3.7k schema tokens

Comprehensive 20-Task Benchmark Suite Evaluation (Budget K=5)

Wilcoxon p = 0.0455 < 0.05

Pipeline Configuration Mode	Exposed Tokens	Token Red. %	Precision @ 5	Recall @ 5	F1 Score	Success Rate
Mode A: Naive Bypass Baseline (Load-All)	7,450.0	0.0%	0.0286	1.0000	0.0554	90.0%
Ablation B1: No-ProjMatch ($w_2 = 0$)	699.0	90.6%	0.1900	0.7500	0.2976	80.0%
Ablation B4: No-Top-K (Unbounded Return)	621.5	91.7%	0.1700	0.7000	0.2690	80.0%
Mode B: Full System (Equal Weights)	725.1	90.3%	0.2100	0.8000	0.3262	85.0%
Tuned System (LOO-CV Calibrated w*)	**725.1**	90.3%	**0.2300**	**0.8500**	**0.3547**	**85.0%**

SUMMARY & IMPACT

Slide 7 — Conclusion: Impact & Contributions

SYNTHESIS

1. Solved the Context Dilemma

Eliminated the forced compromise between expanding agent tool capabilities and preserving scarce prompt window space.

- Reduces schema overhead by **90.3% to 96.7%** across tasks.
- Maintains **85.0% task completion success rate**.

2. Massive Economic ROI

Direct commercial benefits for engineering teams deploying autonomous coding assistants:

- Saves over **420,000 input tokens** per developer in a 20-turn session.
- Over **$29,000 USD / year** direct API token savings for 50 engineers.
- Runs on standard CPU with zero GPU cost.

3. Production Ecosystem Value

A future-proof middleware layer for the next generation of agent systems:

- **Plug-and-Play:** Compatible with Claude Code, Cursor, and OpenCode with zero agent code changes.
- **Fault-Tolerant:** Dual failovers (Postgres/SQLite & Chroma/TF-IDF).
- Ready for fleets of 100+ specialized servers.

MCP Orchestrator Presentation Major Findings Enterprise Value Slide 07 / 08

ACADEMIC & TECHNICAL CITATIONS

Slide 8 — References & Scientific Literature

BIBLIOGRAPHY

📚 Foundational Research Literature

- **Toolformer:** Schick, T., et al. (2023). Self-supervised tool call generation.
- **Gorilla:** Patil, S. G., et al. (2023). Documents tool selection degradation over large API sets.
- **ToolBench:** Qin, Y., et al. (2023). Multi-turn execution evaluation benchmarks.
- **AnyTool:** Du, Y., et al. (2024). Hierarchical coarse-to-fine API selection.
- **Lost in the Middle:** Liu, N. F., et al. (2023). U-shaped performance drop in large contexts.

🛠️ Protocols, Libraries & Specs

- **Model Context Protocol (MCP):** Anthropic PBC. (2024). JSON-RPC 2.0 over stdio and SSE.
- **Sentence-BERT:** Reimers & Gurevych (2019). Dense MiniLM embeddings.
- **Thompson Sampling:** Agrawal & Goyal (2012). Bayesian Beta-Binomial scoring.
- **LangGraph:** LangChain AI. (2024). Multi-agent state machine.

Open Source Repository & Gateway:
`github.com/PatilSharvil/MCP-Orchestrator`